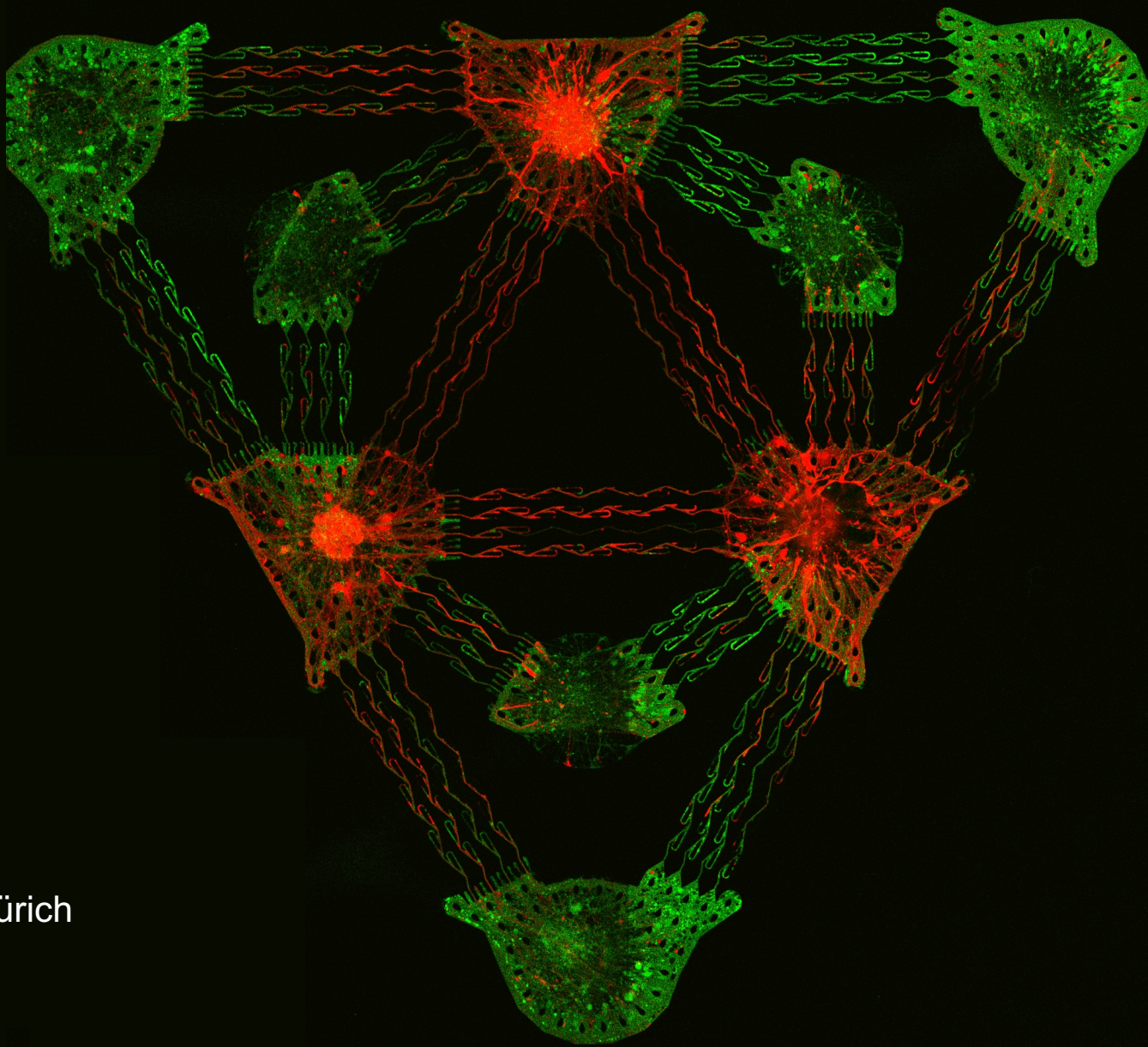




Introduction to the final task

P&S 2026

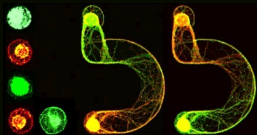
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What have you learnt so far?

Neuroscience

What are neurons and how do they behave?
What are different ways to study and observe neural systems?
What is bottom-up neuroscience?
How can different small neural circuits enable elementary information-processing functions, and how large-scale organization leads to complex behavior?

Neuroengineering

How physical and chemical patterning tools can constrain connectivity and impose specific topologies?
Principles of extracellular electrophysiology, HD-MEA technology.
How can engineered networks help us understand the flow of information, computation, and disease in the brain?

Machine learning

How does machine learning differ from other algorithms?
What are different types of machine learning?
How to identify and avoid overfitting?

Experiments with cells

How to work aseptically?
How to pipette?
How to identify and avoid contamination?

“Just as a small set of recurring words builds all of language, a small set of recurring connectivity patterns (motifs) builds the vast diversity of neural circuits.”

- **Feedforward excitation:** signal propagation and transformation
- **Feedforward/feedback inhibition:** gain control, temporal sharpening, E/I balance
- **Lateral inhibition:** contrast enhancement, winner-take-all selection
- **Mutual inhibition:** rhythmogenesis, state switching
- **Recurrent excitatory loops:** reverberation, memory, amplification

MicroNetwork Motifs

A. Feedforward excitation



D. Lateral inhibition



B. Feedforward inhibition



E. Feedback/Recurrent inhibition



F. Feedback/Recurrent excitation



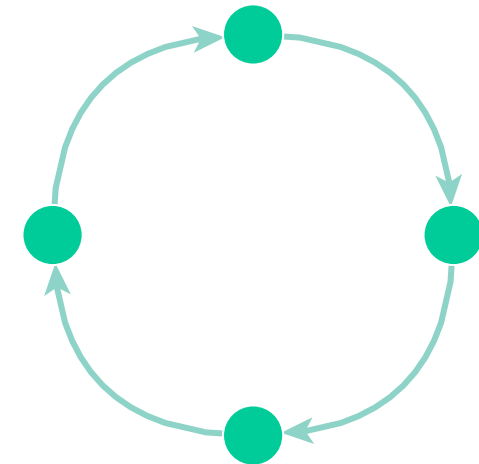
C. Convergence/divergence



— Excitation
— Inhibition

Figure 6

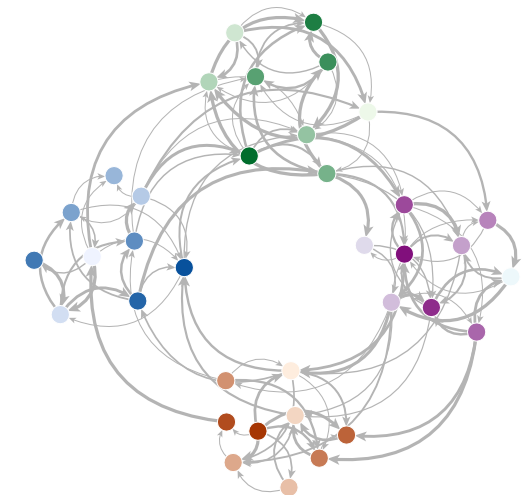
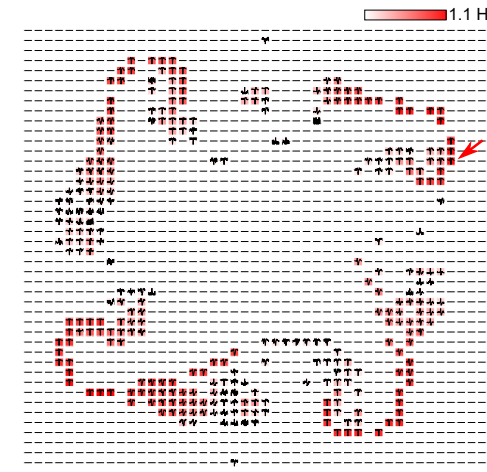
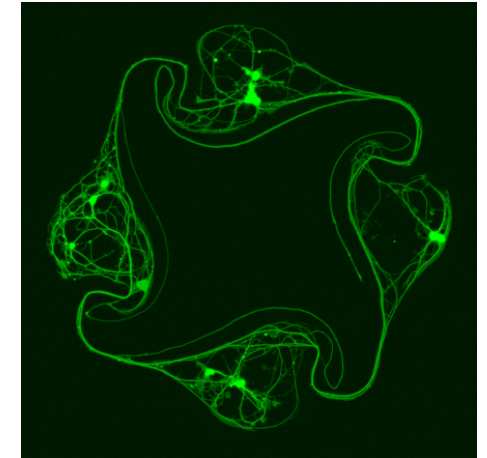
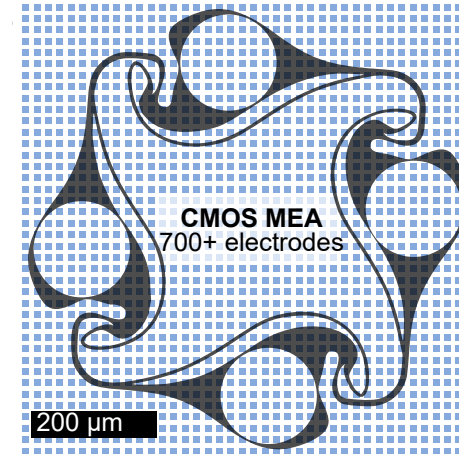
- **Recurrent excitatory loop:** a circuit in which excitatory neurons form a closed cycle of connections, so that activity can propagate from a neuron back to itself through one or more intermediary excitatory neurons.
- **Relevance within larger brain organization:** CA3 recurrent collaterals in the hippocampus supporting pattern completion and memory retrieval; thalamo-cortical loops supporting arousal, attention, and conscious perception; prefrontal cortex recurrent circuits maintaining persistent activity for working memory; layer 2/3 recurrent excitation in the neocortex amplifying and sharpening sensory representations...
- **Computation:** larger number of attractors and more complex state space, stabilization of equilibria against perturbations to initial conditions, signal amplification, activity reverberation after stimulus, oscillation.



Excitatory feedback cycle (clockwise)

Engineering excitatory loops

- We use PDMS microstructure that enforces feedback motif topology.
- The microstructure is positioned on CMOS MEA which allows to record and stimulate at high spatio-temporal resolution.
- We can infer the effective connectivity from recorded activity to confirm that the physical topology imposed by the PDMS translates into functional directed connectivity.
- By stimulating individual electrodes and recording the network response across all electrodes, we can characterize the input-output transformation of the engineered motif.



Why study stimulated activity?

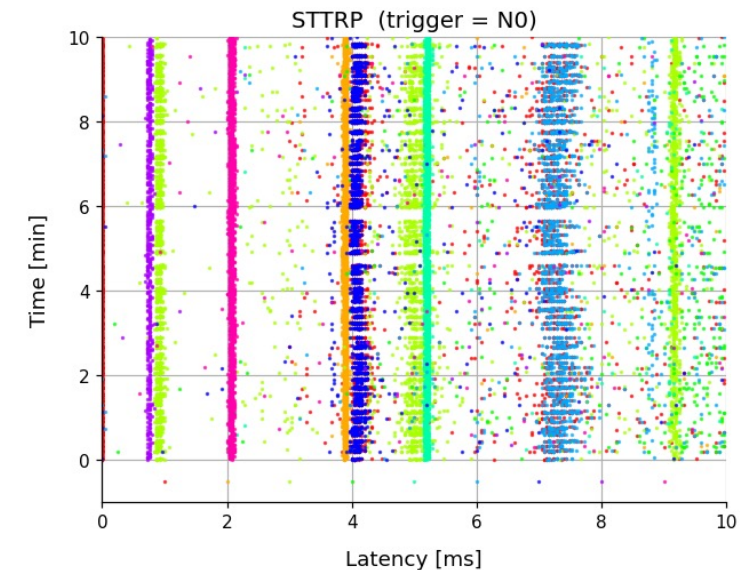
Spontaneous activity:

- Reflects network's preferred states ("attractors").
- Functional connectivity: correlated spontaneous activity is the basis for estimating who is connected to whom
- **Intrinsic dynamical repertoire**: bursting patterns, oscillatory modes, synchronization structure and how it spontaneously emerges from the network structure.

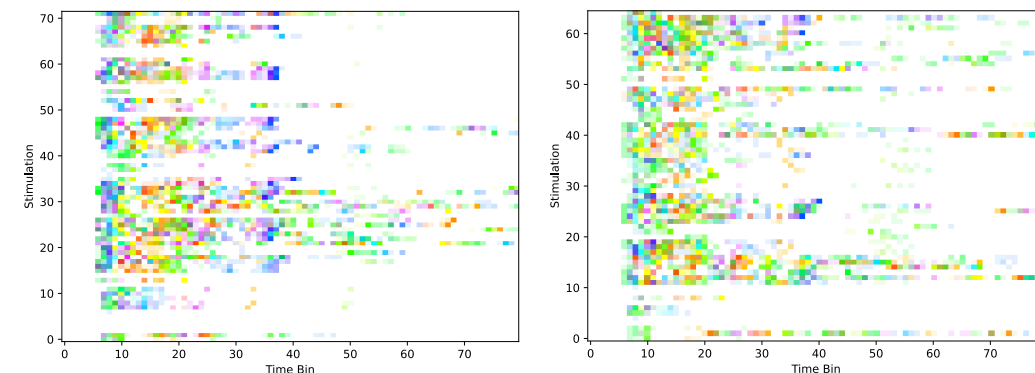
Stimulated activity:

- **Probing the input-output function**: How does the circuit filter, amplify, or transform different input patterns?
- **State-dependence**: internal degrees of freedom revealed by diversity of responses depending on the network's current state.
- **Stimulus specificity**: Does the response depend on which location is stimulated, or at what frequency?

STTRP of spontaneous activity



Post-stimulus response to two different stimuli

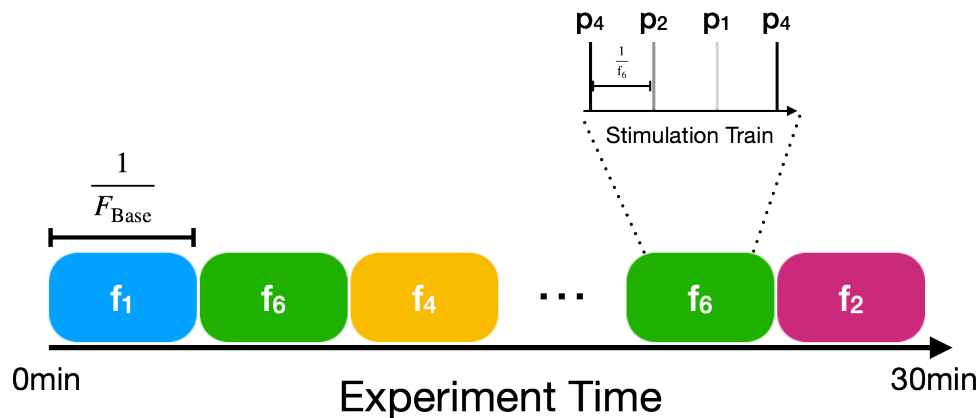
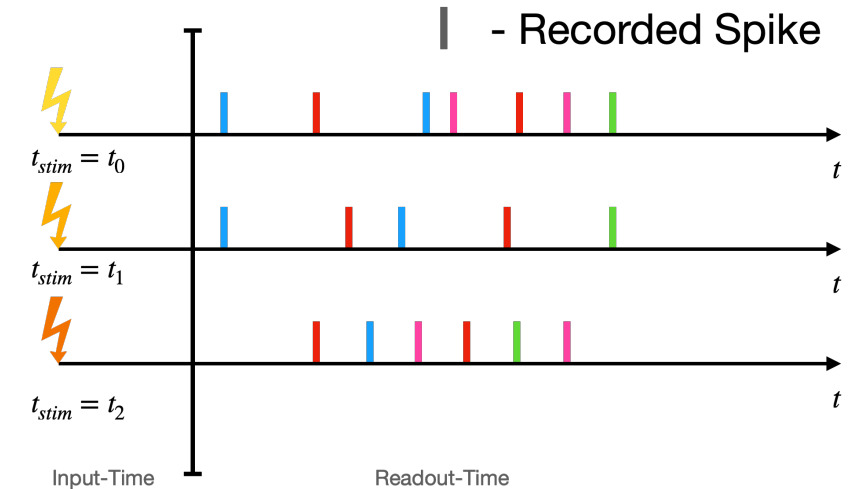
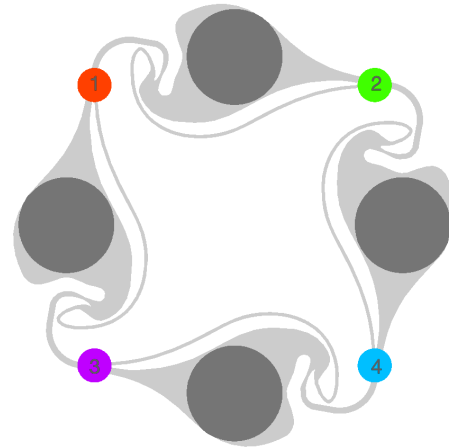


Final Task: data

Stimulated data:

Data comes in tuples of (x, y, z) , where:

- $z \in (\{0, 1, \dots, 15\}, \mathbb{R}^1)$ is stimulation parameters: stimulation pattern and stimulation frequency.
- $y \in \{0, 1, \dots, 15\}$ is the stimulation pattern
- $x \in \{0, 1\}^{N \times T}$, where $N \approx 200$ and $T = 80$ is the number of time bins in the readout time. After-stimulus response is recorded for 20ms at 20kHz (time bin of $250\mu\text{s}$).



- Base frequency F_{base} : how often you change stimulation train
- Stimulation train frequency f_i : with what frequency you apply different patterns p within the stimulation train

Task 1 — Decoding: What stimulation produced this response?

Given a network response $\mathbf{x}$, predict the stimulation pattern $\mathbf{y}$ that caused it.

This is a classification problem — you're building a decoder that reads out the stimulus identity from the population response.

Questions to explore: How does decoding accuracy depend on the temporal resolution of your binned response? Can a model trained at one sampling frequency generalize to another?

Task 2 — Encoding: What response will this stimulation produce?

Given stimulation parameters $\mathbf{z}$, predict the network response $\mathbf{x}$ that will result.

This is the harder and more scientifically interesting direction — you're learning the input-output transformation of the circuit itself. Your model can be discriminative or generative.

The key question: can the model generalize to stimulation parameters it hasn't seen during training?

Timeline:

- Today — you prepare the cultures that will generate your data.
- Next week — you will learn more advanced machine learning tools necessary for Task 2.
- 14/4/2026 — once mature, you will perform stimulation experiments.
- 21/4/2026-19/5/2026 — building ML models.
- Final week — we give you a new, unseen dataset. You run your pipeline on it and present the results.

Test data while the cells are maturing

We have some initial recordings for you to get an idea about the data.

Final_Task on git contains a visualization script for the data and an example of the decoding task.

The data you look at you can choose through four variables:

```
# Load the specific network here
```

```
Network = 3
```

← Network to load, there is 1-4

```
DIV = 28
```

← DIV of the recording (currently, only 28)

```
group_data = False
```

← This you will set True in the future to load your own data

```
test_data = False
```

← This will be needed in the end for the final task challenge

Each group holds a **presentation of 10min**

Explain your **approach**, no need to reexplain stuff from the lectures

1. What method did you use? What is the key mechanism?
2. What assumptions did you make? How are they reflected in your model?
3. How does your algorithm learn over time?

Focus on the **results**

1. How did your method perform? Why did it work/fail?
2. How did you check whether elements of your model meaningfully contribute?

You can only win

1. Machine Learning, neuroscience, neuroengineering knowledge: *priceless*
2. Friendships along the way: *priceless*
3. A voucher for the best performing groups



30 CHF pp



15 CHF pp

Depending on your
rank on the reward ranking.



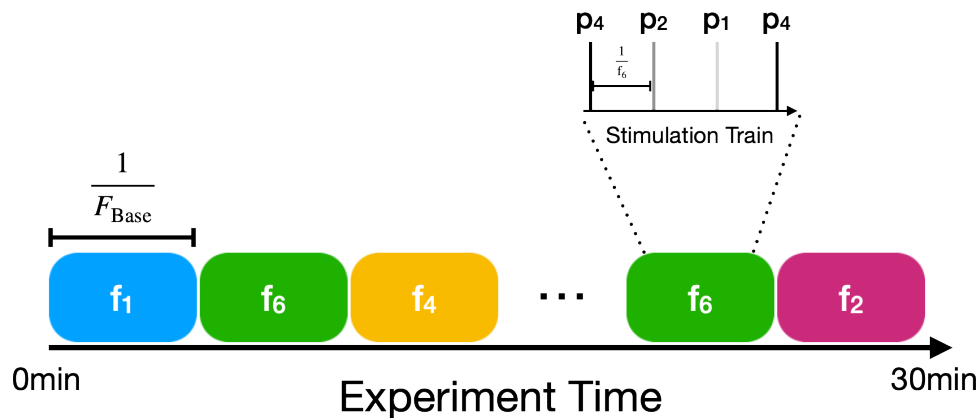
+15 CHF pp

For the most creative
solution.

Until next session, please do the torch task in `pands_ibnnwml/3_NN_Exercises`

Also think about a stimulation protocol for collecting your data. What you can tune:

1. Base Frequency F_{base} (How often the frequency of the stimulation train is changed): min: 0.4Hz, max: 40Hz
2. Range of frequencies f applied: min: F_{base} , max: 40Hz
3. Step size (from range of frequencies, with what stepsize you sweep): float
4. Number of repeats (how many times you repeat the stimulation train of specific f_i): Integer



- Base frequency F_{base} : how often you change stimulation train
- Stimulation train frequency f_i : with what frequency you apply different patterns p within the stimulation train

Suggest two sets of stimulation protocols, each of 30min length. Send your proposals by **mail one per group**. Deadline for this task is 2.4.2026